

COMPUTER SCIENCE PATHWAYS: FROM IGCSE TO UNIVERSITY AND CAREER

计算机科学升学与职业发展路径指南

Prepared for: VPA Shenzhen

Document 7 of 8 | Supporting Research Document

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and A-Level/IB Admissions to Chinese Universities

SECTION 1: THE IGCSE → A-LEVEL → UNIVERSITY PIPELINE

IGCSE → A-Level → 大学升学路径

1.1 IGCSE COMPUTER SCIENCE (0478) AT VPA SHENZHEN

IGCSE 计算机科学 (0478) 在 VPA 深圳

The International General Certificate of Secondary Education (IGCSE) in Computer Science (Syllabus 0478) provides the foundational academic qualification for students beginning their computer science journey at VPA Shenzhen. The course covers ten essential topic areas:

- Topic 1: Data Representation (数据表示) — binary, hex, sound, images
- Topic 2: Data Transmission (数据传输) — protocols, error detection
- Topic 3: Hardware (硬件) — CPU, memory, storage, logic gates
- Topic 4: Software (软件) — OS, utility, high-level vs low-level
- Topic 5: The Internet & Its Uses (互联网及其应用) — security, www
- Topic 6: Automated & Emerging Technology (自动化与新兴技术)
- Topic 7: Algorithm Design (算法设计) — pseudocode, flowcharts
- Topic 8: Programming (编程) — Python fundamentals, problem-solving
- Topic 9: Databases (数据库) — SQL, flat-file vs relational
- Topic 10: Boolean Logic (布尔逻辑) — logic gates, truth tables

Assessment: Two examination papers (Theory + Problem-solving) plus a practical programming pre-release task.

1.2 PROGRESSION TO A-LEVEL COMPUTER SCIENCE (9618)

升读 A-Level 计算机科学 (9618)

Students who complete IGCSE Computer Science at VPA progress naturally to A-Level Computer Science (Syllabus 9618), which deepens and extends all IGCSE topics:



A-Level Paper	Content Builds From IGCSE	
Paper 1: Theory Fundamentals	Data representation, hardware,	
(理论基础)	software, Boolean logic, networking	
	— all IGCSE topics at greater depth	
Paper 2: Fundamental Problem-	Algorithms, programming (Python),	
solving & Programming (编程)	data structures, OOP fundamentals	
Paper 3: Advanced Theory (A2)	Advanced computer architecture,	
(高级理论)	logic circuits, parallel processing	
Paper 4: Practical Programming	Practical on-screen programming	
(实践编程)	project, building on IGCSE skills	

1.3 THE 4-SEMESTER PROGRAMME AT VPA: BUILT SEQUENTIALLY

VPA 四学期课程：循序渐进的体系设计

The IGCSE Computer Science programme is designed for delivery across four semesters (Year 9–Year 10), designed as a deliberate staircase:

Semester 1 (Year 9): FOUNDATIONS (基础)

- Binary, hex, hardware fundamentals, introduction to Python
- Students acquire the "language" of computing

Semester 2 (Year 9): STRUCTURES (结构)

- Algorithms, flowcharts, pseudocode, database concepts, SQL
- Students begin solving structured problems

Semester 3 (Year 10): INTEGRATION (综合)

- Networks, security, Internet protocols, Boolean logic, advanced SQL
- Students connect hardware, software, and network concepts

Semester 4 (Year 10): MASTERY & EXAM PREPARATION (精通与备考)

- Mock examinations (5-mock system), programming projects, emerging technology (AI, robotics)
- Students develop exam confidence and applied capability

This 4-semester architecture ensures that by the time students enter A-Level, they have mastered all foundational concepts TWICE — once at IGCSE level, then again through mock-exam reinforcement — giving them a significant head start over peers who begin CS at A-Level.

SECTION 2: CHINA'S HIGHER EDUCATION REVOLUTION

中国高等教育的革命性变革

本节所有数据均来自中华人民共和国教育部官方公告、国务院文件及权威媒体报道。All data cited from official PRC Ministry of Education

announcements and verified sources.

2.1 THE 20% OPTIMISATION TARGET: ACHIEVED AHEAD OF SCHEDULE

20%优化调整目标：提前完成

In March 2023, China's Ministry of Education (MOE) issued its most significant higher education reform policy in decades:

《普通高等教育学科专业设置调整优化改革方案》	
General Higher Education Subject and Programme Setup Adjustment and Optimisation Reform Plan	
Document Reference: 教高〔2023〕1号	
Key Target: "By 2025, optimise and adjust approximately 20% of higher education subject and programme placements"	
(到2025年，优化调整高校20%左右学科专业布点)	

This target was ACHIEVED AHEAD OF SCHEDULE. In August 2025, Vice-Minister of Education Wu Yan (吴岩) announced that the 20% optimisation goal had been met, describing the "intensity of the adjustment as unprecedented" (调整力度之大，前所未有).
Source: MOE news briefing, August 2025; confirmed by Luan Zongtao (栾宗涛), Deputy Director of MOE Department of Degree Management and Postgraduate Education, 28 August 2025.

THE SCALE OF CHANGE:

2024 SINGLE-YEAR FIGURES (2024 年单年度数据)	
• 1,428 undergraduate programmes REVOKED (撤销)	
• 2,220 undergraduate programmes SUSPENDED (停招)	
• Combined: 3,648 programmes eliminated in one cycle	
• 1,839 NEW programmes added (primarily STEM/AI)	
Source: MOE 2024 Annual Results, 教高函〔2025〕3号, 22 April 2025	
5-YEAR CUMULATIVE FIGURES (2020-2024 累计数据)	
• 5,345+ undergraduate programmes revoked	
• 4,000+ additional programme placements exited via suspension	

Source: Changjiang Daily / 长江日报	
2-YEAR PERIOD (2023-2025)	
• Undergraduate: 6,638 programmes cancelled or suspended	
• Undergraduate: 3,715 new programmes added	
Source: MOE, August 2025	

2.2 FIELDS BEING PRIORITISED: THE "NEW QUALITY PRODUCTIVE FORCES"

优先发展领域：“新质生产力” (Xīn Zhì Shēngchǎn Lì)

The reform is explicitly designed to serve China's "New Quality Productive Forces" (新质生产力) framework, promoted by President Xi Jinping since 2023. This concept refers to economic growth driven by innovation, emerging technologies, and advanced productivity rather than traditional labour and capital inputs.

PRIORITY FIELDS FOR NEW PROGRAMME EXPANSION (官方优先发展领域):

English	Chinese	
Integrated Circuits	集成电路 (Jíchéng Diànlù)	
Artificial Intelligence	人工智能 (Réngōng Zhìnéng)	
Biotechnology	生物技术 (Shēngwù Jìshù)	
Quantum Science & Technology	量子科技 (Liàngǐ Kējì)	
New Energy	新能源 (Xīn Néngyuán)	
New Materials	新材料 (Xīn Cáiliào)	
Low-Altitude Economy / UAVs	低空经济 (Dīkōng Jīngjì)	
Embodied Intelligence	具身智能 (Jùshēn Zhìnéng)	
Autonomous Driving	自动驾驶 (Zìdòng Jiàoshǐ)	
Financial Technology	金融科技 (Jīnróng Kējì)	

Source: MOE Notice on 2025 Annual Undergraduate Programme Setup Work, 《关于开展 2025 年度普通高等学校本科专业设置工作的通知》, 教高司函〔2025〕3 号, 23 June 2025.

NEW "GREEN CHANNEL" MECHANISM (超常设置机制):

A new "extraordinary setup mechanism" allows IMMEDIATE approval for programmes responding to urgent national strategic needs, bypassing the normal annual application cycle. First used in 2025: 6 universities

approved to set up Low-Altitude Technology and Engineering programmes mid-cycle.

Source: MOE 2025 Setup Work Notice, 30 June 2025.

2.3 THREE-YEAR ACTION PLAN (2025-2027)

三年行动计划（2025-2027 年）

Announced 28 August 2025 by the Central Education Work Leading Group of the CPC Central Committee:

Key elements:

- "Extraordinary deployment initiative" (超常布局举措) for urgently needed disciplines targeting strategic emerging industries
- AI-powered big data platform to track talent supply/demand with "precision" and provide early warnings on programmes falling short of industry needs
 - Accelerating AI empowerment of education and teaching
 - Establishing exemplary interdisciplinary centres
 - Creating national innovation institutes for outstanding engineers

Source: University World News, 12 September 2025; Global Times, 25 August 2025.

2.4 KEY STATISTICS: THE TALENT GAP CRISIS

关键数据：人才缺口危机

SEMICONDUCTOR SHORTAGE (半导体人才缺口):

2025 projected shortage: 300,000–350,000 workers	
(中国半导体产业人才缺口预计将达 30 万–35 万人)	
Advanced process R&D talent gap alone: exceeding 120,000	
Packaging and testing engineers gap: exceeding 20,000	
Graduate pipeline problem:	
• 210,000 students graduated from IC-related programmes in 2020	
• ONLY 13.77% (fewer than 30,000) actually entered the IC industry	
Source: MIIT 《中国集成电路产业人才发展报告》	
China Semiconductor Industry Association	

AI TALENT CRISIS (人工智能人才危机):

AI supply-demand ratio: 0.39	
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	(每十个岗位空缺中合格的候选人才不足四人)	
	— Meaning: for every 10 open AI positions, fewer than 4 qualified	
	candidates exist	
	AI starting salary average: RMB 18,592/month (~USD 2,600)	
	— Highest among all fields	
	AI architect positions: Monthly salary exceeding RMB 42,000	
	AI researchers in China: 53,000+	
	AI companies: 4,500+	
	Source: NDRC Expert Article (May 2026);	
	Zhaopin 智联招聘 "2025 中国半导体产业人才报告"	

UNIVERSITY EXPANSION (高校扩招):

	535+ Chinese universities now offer AI undergraduate programmes	
	(对比美国约 14 所)	
	43 specialised AI schools and research institutes established	
	since 2017	
	Notable developments:	
	• Tsinghua University: College of AI established April 2024,	
	led by Turing Award winner Andrew Yao (姚期智)	
	• Shanghai Jiao Tong University: School of AI established 2024,	
	added 150 spots in cutting-edge tech for 2025	
	• Peking University: Added 150 spots in strategic areas for 2025	
	• Zhejiang University: Introductory AI class MANDATORY for all	
	undergraduates from 2024	
	• Fudan University: 116 "AI-BEST" courses, 13,000+ students	
	enrolled in AI-related courses	
	Source: University World News; Global Times;	
	MOE English Portal, 31 March 2025	

SECTION 3: CHINESE UNIVERSITY ADMISSIONS FOR COMPUTER SCIENCE

中国大学计算机专业招生

3.1 ALL C9 LEAGUE UNIVERSITIES ACCEPT A-LEVEL/IB

九校联盟（C9）全部接受 A-Level/IB 成绩

The C9 League (九校联盟) is China's most prestigious university

alliance. ALL nine universities accept A-Level and IB qualifications for international student admissions to Computer Science, AI, and Software Engineering programmes:

University (English)	Chinese Name	CS/AI Programme Notes
1. Tsinghua University	清华大学	College of AI est. 2024; Andrew Yao; HSK 5 required
2. Peking University	北京大学	New Engineering & Info Science Divisions established 2025
3. Fudan University	复旦大学	116 AI-BEST courses; 41 "X+AI" dual-degree
4. Shanghai Jiao Tong University (SJTU)	上海交通大学	"Special zone" AI college; 150 new spots
5. Zhejiang University	浙江大学	Mandatory AI for all undergraduates; strong CS/AI programme
6. University of Science and Technology (USTC)	中国科学技术大学	Research-intensive; CSC scholarships
7. Nanjing University	南京大学	Top 10 AI nationally
8. Xi'an Jiaotong Univ.	西安交通大学	Engineering-focused
9. Harbin Institute of Technology (HIT)	哈尔滨工业大学	A+ AI rating (ARWU 2024); strong CS/SE

Source: Individual university international admissions websites; Cambridge International "Moving on to University" 2024.

3.2 SCHOLARSHIP OPPORTUNITIES

奖学金机会

CHINESE GOVERNMENT SCHOLARSHIP (CSC — 中国政府奖学金):

Coverage: Full tuition waiver + accommodation + medical insurance
+ monthly stipend
Stipend amounts:
• Undergraduate: ~2,500 RMB/month

• Master's: 3,000 RMB/month	
• PhD: 3,500 RMB/month	
Application: https://studyinchina.csc.edu.cn	
Timeline: October–February annually	
Also available:	
• Tsinghua University Institutional Scholarship (full/half/partial)	
• Beijing Government Scholarship	
• USTC Fellowship	
• Silk Road Programme (Belt and Road STEM focus)	

Source: Chinese Government Scholarship portal; MOE Document No. 12 [2020].

3.3 THE IDEAL A-LEVEL COMBINATION FOR CHINESE CS PROGRAMMES

申请中国 CS 专业的理想 A-Level 科目组合

Chinese universities value A-Level/IB Computer Science as a STEM subject for admission. The ideal combination is:

RECOMMENDED COMBINATION (推荐组合):

Mathematics (数学)	← ESSENTIAL, typically A or above
Physics (物理)	← STRONGLY PREFERRED
Computer Science (计算机科学)	← Viewed favourably, demonstrates commitment to the field
Typical offers for C9 universities: AAB–AAA including Mathematics	
For IB applicants: 34–42 points overall, HL Mathematics essential	

Source: Tsinghua University International Admissions; Peking University International Student Admissions; Cambridge International 2024.

3.4 COMPARISON: A-LEVEL CS vs GAOKAO CS

A-Level 计算机科学 vs 高考计算机科学

CRITICAL FINDING: The Gaokao does NOT include Computer Science as a national subject. China's national university entrance examination has NO dedicated computer science component for the vast majority of provinces.

Aspect	Gaokao System	A-Level / IB System
National CS subject?	NO — not offered	YES — full CS

	nationally	syllabus	
Zhejiang exception	"Technology" (技术)	N/A	
	includes IT comp-		
	onent, but this is		
	UNIQUE to Zhejiang		
Programming depth	Basic (Zhejiang)	Substantial	
	Not covered	(IGCSE→A-Level/IB	
	(other provinces)	builds progressively)	
Algorithms	Basic	Extensive	
Database/SQL	Not covered	Covered thoroughly	
OOP	Not covered	A-Level (A2),	
		IB (SL+HL)	
Boolean logic	Basic	Extensive	
University readiness	Zhejiang students	Strong foundation	
	show advantage,	for all CS programmes	
	but only in one	globally	
	province		

CONCLUSION: Students with A-Level/IB Computer Science enter university with a SUBSTANTIALLY STRONGER foundation than Gaokao students from provinces without a technology elective. Even compared to Zhejiang's "Technology" subject, A-Level/IB CS offers significantly greater depth in programming, algorithms, databases, and object-oriented design.

Source: Zhejiang Provincial Education Examination Authority;
<https://m.gaokao.cn/zd/25424>

SECTION 4: GLOBAL UNIVERSITY PATHWAYS

全球大学升学路径

A-Level Computer Science (9618) is recognised by leading universities worldwide. Below are top destinations for VPA graduates:

4.1 UNITED KINGDOM (英国)

University	CS Programme Highlights	
University of Oxford (牛津大学)	BA/MCompSci; Computer Science and Mathematics overlap; Tutorial system	
University of Cambridge (剑桥大学)	BA Computer Science; strong theory; Natural Sciences pathway available	

Imperial College London	BEng/MEng Computing; AI/ML special-	
(帝国理工学院)	isation; integrated year in industry	
UCL (London)	BSc Computer Science; strong research	
(伦敦大学学院)	output; Iggy test required	
University of Manchester	BSc CS; birthplace of modern compu-	
(曼彻斯特大学)	ting (Manchester Baby)	
University of Edinburgh	BSc/MInf Informatics; strong AI/ML	
(爱丁堡大学)	research; 4-year Scottish system	

4.2 UNITED STATES (美国)

University	CS Programme Highlights
MIT	EECS (Electrical Eng & CS); world's
(麻省理工学院)	top-ranked; requires SAT/ACT + TOEFL
Stanford University	BS Computer Science; Silicon Valley
(斯坦福大学)	proximity; entrepreneurial culture
Carnegie Mellon (CMU)	SCS (School of Computer Science);
(卡内基梅隆大学)	multiple CS majors; cross-disciplinary
Georgia Tech	BS CS; strong industry connections;
(佐治亚理工学院)	Atlanta tech hub; excellent ROI

4.3 AUSTRALIA & ASIA-PACIFIC (澳大利亚及亚太地区)

University	CS Programme Highlights
University of Melbourne	BSci Computing & Software Systems;
(墨尔本大学)	strong graduate outcomes
Australian National Univ.	BAdvComp(Hons); research-intensive;
(澳洲国立大学)	Canberra location
University of Sydney	BAdvComputing; AI/ML specialisations
(悉尼大学)	available
HKU (Hong Kong)	BEng CS; Asia's oldest university;
(香港大学)	strong industry links
HKUST (Hong Kong)	BEng/BSc CS; strong AI/ML research;
(香港科技大学)	close to Shenzhen
CUHK (Hong Kong)	BEng CS; Chinese-English bilingual;
(香港中文大学)	strong theoretical foundation

NUS (Singapore)	BComp CS; consistently Asia's #1;	
(新加坡国立大学)	strong industry partnerships	
NTU (Singapore)	BEng CS; rapidly rising rankings;	
(南洋理工大学)	strong AI research	

Note: A-Level grades are universally accepted at all institutions listed. A-Level Computer Science counts as a rigorous STEM subject.

Typical offers: A*AA–AAA for UK Russell Group; AP/ SAT Subject Tests additionally useful for US applications.

SECTION 5: CAREER PATHWAYS IN CHINA

中国职业发展路径

China's technology sector offers exceptional career prospects for graduates with internationally recognised CS qualifications. The government's strategic prioritisation of AI, semiconductors, and software means demand for talent will remain strong for decades.

5.1 SOFTWARE ENGINEERING (软件工程)

Tencent (腾讯) — Shenzhen HQ	
Roles: Backend/frontend engineer, WeChat development, gaming	
Salary: New graduates 25,000–35,000 RMB/month	
Alibaba (阿里巴巴) — Hangzhou HQ	
Roles: Cloud computing, e-commerce platforms, data engineering	
Salary: Competitive; strong stock options	
ByteDance (字节跳动) — Beijing HQ	
Roles: TikTok/Douyin algorithm engineer, recommendation systems	
Salary: Among highest in industry for AI roles	
Huawei (华为) — Shenzhen HQ	
Roles: 5G, telecommunications, consumer electronics, HarmonyOS	
Salary: Strong; "Genius Youth" programme pays 1M+ RMB/year	
DJI (大疆创新) — Shenzhen HQ	
Roles: Drone software, computer vision, autonomous flight	
Salary: Premium for CV/robotics specialists	

5.2 AI / MACHINE LEARNING (人工智能/机器学习)

China's AI sector has a supply-demand ratio of just 0.39 — fewer

than 4 qualified candidates for every 10 positions. Average starting salary: RMB 18,592/month (highest of all fields).

Key employers: Baidu AI, Alibaba DAMO Academy, Tencent AI Lab, SenseTime, Megvii, iFLYTEK, Horizon Robotics.

Source: NDRC Expert Article, May 2026.

5.3 SEMICONDUCTOR INDUSTRY (半导体产业)

With a talent gap of 300,000–350,000 workers, China's semiconductor sector offers massive opportunity:

SMIC (中芯国际) — China's largest chip foundry	
Roles: Process engineer, device physics, yield optimisation	
Hua Hong Semiconductor (华虹半导体)	
Roles: Power IC design, analog/mixed-signal engineering	
Regional opportunity:	
• Shenzhen: Strongest attraction for ALL experience levels	
• Shanghai: 10,000-level semiconductor talent gap	
• Suzhou: Second-place for experienced talent	
• Hefei: Offering PhDs RMB 600,000 housing allowance	

Source: MIIT; 2025 Suzhou Semiconductor Talent Gap Report.

5.4 CYBERSECURITY (网络安全)

National security priority under China's Cybersecurity Law (2017). Government, finance, and critical infrastructure sectors all require certified security professionals. Graduates with international security qualifications (CISSP, CEH) command premium salaries.

5.5 FINTECH (金融科技)

Ant Group (蚂蚁集团), WeChat Pay (微信支付), and JD Finance offer roles in blockchain, risk modelling, algorithmic trading, and payment systems engineering. Shenzhen and Hangzhou are primary hubs.

5.6 GAMING INDUSTRY (游戏产业)

China is the world's largest gaming market by revenue. Tencent Games, NetEase, miHoYo (Genshin Impact), and 莉莉游戏 hire software engineers, graphics programmers, and AI specialists for game development.

5.7 SHENZHEN: CHINA'S SILICON VALLEY

深圳：中国硅谷

VPA Shenzhen students are positioned at the epicentre of China's technology revolution. Shenzhen is home to:

HEADQUARTERS IN SHENZHEN (深圳总部企业):	
• Tencent (腾讯) — Social media, gaming, fintech, cloud	
• Huawei (华为) — Telecommunications, chips, consumer electronics	
• BYD (比亚迪) — Electric vehicles, batteries, energy storage	
• DJI (大疆创新) — Drones, computer vision, robotics	
• ZTE (中兴通讯) — 5G, telecommunications	
• Ping An (平安) — Insurance, fintech, AI	
• Foxconn (富士康) — Electronics manufacturing	
ADVANTAGES FOR VPA STUDENTS:	
• Proximity to internships and industry mentorship	
• Bilingual skills match Shenzhen's international business environment (bilingual Chinese-English = competitive advantage)	
• Local university partnerships (Shenzhen University, SUSTech, Tsinghua SIGS, PKU Shenzhen)	
• Government incentives for tech talent in Greater Bay Area	

SECTION 6: WHY STARTING AT YEAR 9 MATTERS

为什么从 9 年级开始至关重要

6.1 A 4-YEAR HEAD START (四年领先优势)

Students who begin Computer Science at Year 9 (IGCSE) versus those who start at Year 12 (A-Level) gain a transformative advantage:

VPA STUDENT (IGCSE start, Year 9) vs STANDARD STUDENT	
Year 9: IGCSE CS begins — first exposure to programming, hardware, algorithms, databases, Boolean logic	
标准学生: 尚未接触计算机科学	

Year 10: Complete IGCSE, sit examinations — ALL foundational topics mastered; Python proficiency achieved	
标准学生: 尚未接触计算机科学	
Year 11: Begin A-Level CS — topics revisited at greater depth; Paper 1 and Paper 2 preparation begins	
标准学生: First exposure to CS concepts	
Year 12: A-Level examination year — A2 topics (advanced architecture, OOP) + Paper 4 practical programming	
标准学生: Still building foundational knowledge	
RESULT: VPA student has 4 YEARS of cumulative CS education; standard student has 2 years. The depth, confidence, and fluency gap is substantial and measurable.	

6.2 BILINGUAL FOUNDATION: LEARNING CS IN BOTH LANGUAGES

双语基础：用中英两种语言学习计算机科学

VPA's programme delivers Computer Science content in English (the language of international qualifications) while integrating Chinese terminology. This creates a rare and valuable capability:

- Students read Cambridge syllabi in English → sit English exams
- Students learn Chinese technical vocabulary → prepared for Chinese university programmes (HSK + technical Chinese)
- Bilingual fluency in CS terminology is a significant career advantage in China's tech sector, where documentation, teams, and clients frequently operate across both languages

学生同时掌握英文技术文档阅读能力与中文专业术语，为未来在中国或国际科技行业发展奠定独特的语言优势。

6.3 CULTURAL ANCHORS MAKE ABSTRACT CONCEPTS CONCRETE

文化锚点让抽象概念具体化

VPA's China-contextualised curriculum connects abstract computing concepts to students' lived experience:

Abstract Concept	VPA Cultural Anchor
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Data transmission	WeChat messaging architecture	
(数据传输)	微信消息传输架构	
Cybersecurity	China's Cybersecurity Law, Huawei	
(网络安全)	sanctions case study	
Database systems	Alipay transaction processing	
(数据库系统)	支付宝交易处理系统	
Emerging technology	DJI drones, BYD autonomous driving	
(新兴技术)	大疆无人机, 比亚迪自动驾驶	
Boolean logic	WeChat Pay QR code encoding	
(布尔逻辑)	微信支付二维码编码	

These anchors make abstract concepts memorable and meaningful — students don't just learn WHAT binary is; they understand WHY it matters for the technology they use every day.

6.4 THE 5-MOCK SYSTEM: BUILDING EXAM CONFIDENCE PROGRESSIVELY

五模考试体系：逐步建立考试信心

VPA's assessment architecture includes five mock examinations across the 4-semester programme:

Mock 1 (Semester 2): Diagnostic — identifies gaps early

Mock 2 (Semester 3): Formative — builds exam technique

Mock 3 (Semester 3): Consolidation — strengthens weak areas

Mock 4 (Semester 4): Intensive — simulates full exam conditions

Mock 5 (Semester 4): Final — final confidence builder before IGCSE

通过五次模拟考试，学生逐步掌握考试技巧，消除考试焦虑，在真正的 IGCSE 考试中发挥最佳水平。This progressive approach eliminates exam anxiety and ensures peak performance in the actual IGCSE examinations. The result: VPA students sit their IGCSE Computer Science examinations with the confidence of having been examined FIVE times previously — not just on content, but on time management, question technique, and stress management.

CONCLUSION 结语

The pathway from IGCSE Computer Science at VPA Shenzhen leads directly into the most dynamic and strategically important sector of China's economy. With 535+ universities now offering AI programmes, a semiconductor talent gap of 300,000–350,000 workers, and an AI supply-demand ratio of 0.39, students who build their CS foundations early

— beginning at Year 9, progressing through A-Level, and entering university with 4+ years of cumulative education — are positioned to capture extraordinary opportunities.

China's higher education revolution is not a distant policy debate. It is happening NOW — with real programmes cut, real programmes created, and real scholarships available for international students with the right qualifications. VPA's sequential 4-semester programme, bilingual delivery, and China-contextualised content give its students an unmatched competitive advantage.

在 VPA 深圳，从 9 年级开始的计算机科学教育不仅是一门课程——它是通往中国科技未来的护照。

SOURCES AND CITATIONS 来源与引用

[Official Government Sources]

- MOE Policy Document: 教高〔2023〕1 号 (March 2023)
- MOE 2024 Annual Results: 教高函〔2025〕3 号 (22 April 2025)
- MOE 2025 Setup Notice: 教高司函〔2025〕3 号 (23 June 2025)
- Three-Year Action Plan announcement (28 August 2025)
- 《教育强国建设规划纲要(2024—2035 年)》

[Semiconductor & AI Talent Data]

- MIIT 《中国集成电路产业人才发展报告》
- NDRC Expert Article (May 2026): <https://www.ndrc.gov.cn>
- Zhaopin 智联招聘 "2025 中国半导体产业人才报告"
- China Semiconductor Industry Association

[University Admissions Data]

- Tsinghua University International Admissions
- Peking University International Student Admissions
- Cambridge International: "Moving on to University" (2024)
- Chinese Government Scholarship (CSC): <https://studyinchina.csc.edu.cn>
- Zhejiang Provincial Education Examination Authority

[News Sources]

- University World News (12 September 2025)
- Global Times (25 August 2025)
- Changjiang Daily / 长江日报
- The Paper / 澎湃新闻 (9 May 2025)
- Caixin Weekly / 财新周刊 (23 August 2025)

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